

Utkarsh Anshu Gupta

+91-9221298348 | utkarsh.gupta120902@gmail.com | [linkedin.com/in/utkarsh-gupta1209](https://www.linkedin.com/in/utkarsh-gupta1209) | [UtG1209](#)

EDUCATION

Indian Institute of Technology, Bombay

B.Tech Chemical Engineering, Minor in Computer Science

Mumbai, India

Oct 2020 – May 2024

PUBLICATIONS

Hyperspectral Unmixing Algorithms with Explainability: An Assessment of Geometric and Deep Learning Methods

Utkarsh Anshu Gupta, Soumyashree Kar | Oral Presentation | Accepted, IEEE IGARSS 2026

PROFESSIONAL EXPERIENCE

Air India, Tata Group | Data Scientist | Digital and Technology

Nov 2024 – Present

Enterprise Flight Operations Analytics Chatbot

- Architected and deployed a **GPT-4** powered conversational analytics chatbot on Microsoft Teams using PostgreSQL, Azure OpenAI, and Databricks, enabling natural-language flight operations queries with **~10-second** response time
- Engineered **OWASP**-aligned LLM security guardrails with **65+** security detection patterns for prompt injection and sensitive information disclosure, implementing dual-layer sanitization middleware for **100+** non-technical users
- Optimized database infrastructure through asyncpg connection pooling, **TTL** caching, and intelligent query routing, reducing connection overhead by **60–70%** and offloading **40%** of operational workload to Databricks SQL Warehouse

Crew Feedback Intelligence Pipeline

- Built a NLP pipeline using RoBERTa-MNLI with bi-encoder retrieval and cross-encoder re-ranking, classifying clauses across 15 behavioral dimensions at **97.7%** accuracy, matching frontier model performance at **zero** extra API costing
- Deployed end-to-end MLOps pipeline on Databricks using **PySpark**, **Delta Lake**, and Azure DevOps CI/CD with automated file-arrival triggers, scaling to **100k+** feedback entries & reducing days of manual processing to few hours
- Optimised transformer deployment on Databricks via **float16** precision and batch processing, fitting 355M-parameter RoBERTa-MNLI and **22M**-parameter cross-encoder within DS3_v2 memory constraints as a fully self-hosted solution

Reliance Industries Limited | Technical Intern | CTS Department

May 2023 – Jul 2023

Received a 'Pre - Placement Offer' to join as a full-time employee based on outstanding performance

- Drove a **50%** cryogenic ethylene throughput gain by running sensitivity analysis on temperature and pressure-drop deltas across simulated and actual **ASPEN** data to tune parameters while holding output conditions fixed.
- Analyzed **six** heat exchangers across **250+** simulations, statistically comparing variable profiles to quantify deviation

RESEARCH EXPERIENCE

ML Techniques for Hyperspectral Unmixing

Jun 2025 – Present

Guide: Prof. Soumyashree Kar | CSRE, IIT Bombay

- Developed a CNN+ViT hybrid unmixing architecture combining convolutional feature extraction with a transformer based spectral dependency modeling, achieving RMSE of **0.0131** on Jasper Ridge and **0.0119** on Urban benchmarks
- Designed a DDPM - based generative model in the abundance space to synthetically augment training data, reducing endmember error by **6%** and reconstruction error by **28%** on Cuprite closing **87%** of the gap to published SOTA

Data-Driven Discovery of High-Performing Phosphine Ligands

Aug 2024 – Apr 2025

Guide: Prof. Debabrata Maiti | Chemistry, IIT Bombay

- Trained **190** XGBoost models (median $R^2=0.86$, max $R^2=0.96$) to estimate DFT descriptors from **1024-bit** Morgan fingerprints, enabling accurate modelling of high-dimensional quantum chemical properties for phosphine ligands
- Engineered domain-specific feature selection pipeline via threshold optimisation across all DFT descriptors; applied models to fragment-based generated ligands to predict high reaction yield in cross-electrophile coupling

STARTUP EXPERIENCE

NYX | AI Engineer | IMG – IMG Generation

Jul 2024 – Aug 2024

- Developed an IMG-IMG generation pipeline using **JuggernautV9** to reduce total customer acquisition cost by **30%**
- Preprocessed images using BRIA-RMBG - 1.4 for background segmentation and RealVisXL_V4.0 model for masking

Xequence AI | ML Engineer | Chatbot Development

Jun 2023 – Aug 2023

- Created a end-to-end operational Indian Railways WhatsApp assistant chatbot using **Rapidbott** and OpenAI APIs
- Built a RASA-based ML model to convert speech input into prompts according to a predetermined threshold criteria

TECHNICAL SKILLS

Languages: Python, SQL, MATLAB, LaTeX

Machine Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost, Transformers, OpenCV, SBERT

Cloud & MLOps: Azure, Azure DevOps, Docker, Databricks, PySpark, Delta Lake, PostgreSQL, MongoDB